

Lecture 10

Entity Framework Core Fundamentals

Continuation of the layered Expense Tracker

EF Core is not business logic; it is an Infrastructure tool.

- Introduce DbContext, DbSet, LINQ, tracking and SaveChangesAsync.
- Replace repetitive row mapping without removing layers.
- Use SQLite for persistent local data.

Domain: Expense.

Application: IExpenseRepository and use cases.

Infrastructure: DbContext plus EF repository.

Presentation: input conversion.

Project: DbContextOptions.

`DbContext` is a short-lived unit of work.

`DbSet` is a queryable set.

`ToListAsync` executes SQL.

`AsNoTracking` fits read-only queries.

- Read the runnable CodeExamples project first.
- Run the Project and identify where each responsibility lives.
- Change one implementation detail without breaking the Application layer.
- Write a short explanation of the boundary you protected.

- Apply the same decision to the student's chosen domain.
- Keep a screenshot or short note for the evolution portfolio.
- Use the database-session model and constraints.
- Prepare one demonstration scenario for the final defense.